



Ahmedabad
University

Differentiable Causal Discovery Algorithm with Skeleton Posterior Graph (DCD-SPOT)

Project - Milestone 1

Anant Narayan (AU2420009)

Ankit Kumar (AU2420147)

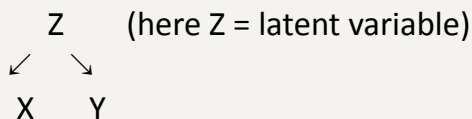
Milan Shah (AU2420195)

Course: CSE516 – Probabilistic Graphical Models

Team: g6_th_apl_cybersecurity_team

What Problem Are We Studying?

- Goal: recover causal structure from observational data
- Challenge: some causally relevant variables may be **unobserved**
- Unobserved variables can create **latent confounding**
- Standard DAG-based methods assume causal sufficiency
- With latent confounders, the observed causal structure is better represented using **ADMGs/MAGs**
- Visual suggestion -



Observed representation = $X \leftrightarrow Y$

Why Existing Differentiable Methods Are Not Enough?

- Differentiable causal discovery made DAG learning tractable through continuous optimization
- But real-world data often contains latent confounders
- Latent confounding requires learning **MAGs**, which are more complex than DAGs
- Existing differentiable MAG methods struggled with larger graphs
- A second problem: using a **single estimated skeleton** can propagate false/missing edges into downstream optimization

→ SPOT identifies this as the central motivation.

From NOTEARS to SPOT

Evolution of Differentiable Causal Discovery

NOTEARS	SPOT
DAG learning	MAG learning
Assumes acyclic directed graph	Handles latent confounding
Continuous optimization	Continuous optimization + posterior guidance
Hard acyclicity constraint	Skeleton posterior + stochastic optimization
Point graph estimate	Probabilistic skeleton information

Project Scope

Our Theoretical AI Focus

Selected Algorithm: SPOT — Skeleton Posterior-guided OpTimization

Methodological Problem: Causal structure learning in the presence of latent confounders

Core Question: How does probabilistic skeleton information affect causal structure learning compared with using a single estimated skeleton?

Scope:

- Study SPOT's mathematical formulation
- Understand its PGM components
- Reproduce/evaluate its causal structure-learning pipeline
- Use the **Sachs benchmark**
- Compare against appropriate causal-discovery baselines
- Analyze performance and limitations

Research Question

Our Main Research Question: Can a posterior distribution over possible causal skeletons improve the accuracy and robustness of differentiable causal discovery under latent confounding?

Supporting questions

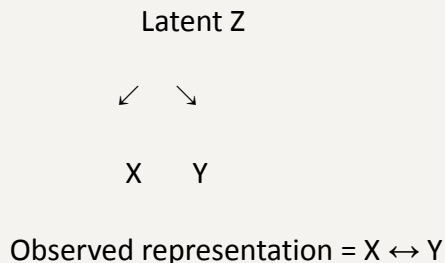
- How accurately can the skeleton posterior be estimated?
- Does posterior-based guidance reduce erroneous/missing edges?
- How does SPOT compare with existing MAG-learning methods?
- How does performance change with graph size and data characteristics?

PGM Representation

Graphical Representation

- **Nodes:** observed variables
- **Latent variables:** unobserved common causes
- **Directed edge $X \rightarrow Y$:** causal influence
- **Bidirected edge $X \leftrightarrow Y$:** latent confounding
- **Skeleton:** undirected structure indicating adjacency
- **MAG:** maximal ancestral graph representing the observed causal structure

Example -



Why this representation?

- DAGs cannot explicitly represent latent confounding among observed variables
- MAGs/ADMGs can encode both directed relationships and latent-confounding relationships

PGM Vertical Mapping

Representation → Learning → Inference

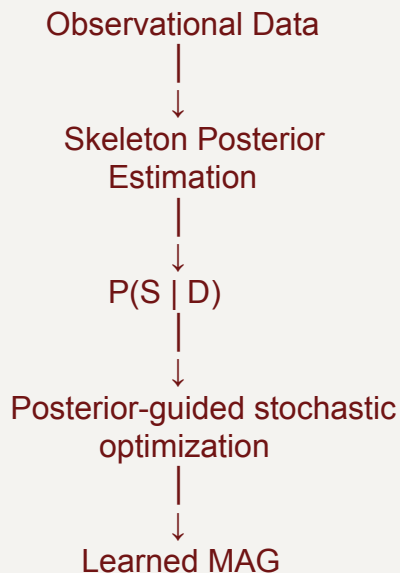
PGM Vertical	Our Project
Representation	ADMG / MAG + skeleton
Learning	Skeleton posterior estimation + differentiable MAG optimization
Inference	Estimate $P(S \mid D)$ and use it to guide causal structure optimization

Core objects

$D = \{x^1, \dots, x^n\}$ $S = \text{skeleton}$ $P(S \mid D)$

SPOT Overview

Two-Phase Framework:



Two major stages

Phase 1 — Skeleton Posterior

- Estimate probability of adjacency
- Uses amortized inference
- Supervised causal learning model

Phase 2 — MAG Optimization

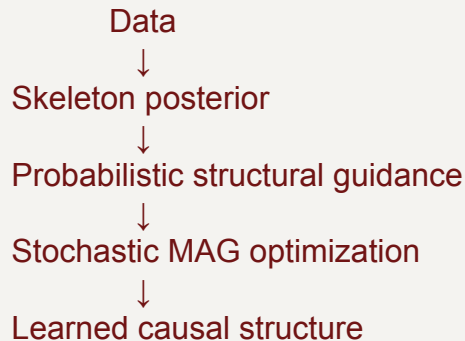
- Use skeleton posterior during optimization
- Stochastically update graph variables
- Reduce unnecessary search space

Posterior-Guided Optimization

Without SPOT:

Data → Differentiable MAG optimization

With SPOT:



Why stochastic guidance?

- Hard skeleton constraints can propagate false negatives
- Posterior allows uncertainty instead of forcing binary decisions
- Search space is reduced
- Optimization can focus on more plausible structures

SPOT replaces deterministic gradient-only updates with posterior-guided stochastic updates.

Assumptions and Scope

SPOT's Main Setting

- Observational data
- Linear Gaussian SCM for the main formulation
- Presence of latent confounders
- No selection bias
- No undirected edges in the ADMG
- Target: causal structure represented through MAG/ADMG

Important limitation

- Real-world data may violate linear-Gaussian assumptions
- Sachs is **non-Gaussian**
- Therefore, Sachs is particularly useful for studying how the method behaves outside ideal assumptions

Sachs Dataset

Real-World Benchmark

Sachs protein-signaling dataset

- Protein-expression measurements
- Human immune-system cells
- **853 observations**
- Variables include:
 - Raf
 - Mek
 - Erk
 - Akt
 - PKA
 - etc.
- Used for causal structure discovery
- Reference biological network available for evaluation

Why Sachs?

- Real-world causal-discovery benchmark
 - Established reference structure
 - Small enough for detailed analysis
 - Used directly in the SPOT paper
- SPOT reports a Sachs skeleton F1 of **0.63**, compared with **0.48 for ABIC**

Baseline Strategy

Primary baseline : ABIC

- Differentiable MAG-learning method
- Directly relevant to latent-confounder setting
- Used as SPOT's main counterpart
- Optimizes directed and bidirected edge structures

Secondary / historical reference : NOTEARS

- Important predecessor in differentiable causal discovery
- Learns DAGs using continuous optimization
- Does not directly model latent-confounded MAGs
- Used to understand the methodological evolution toward SPOT

Classical reference

FCI / RFCI

Constraint-based causal discovery

State-of-the-Art Reference — SPOT

Paper: *Scalable Differentiable Causal Discovery in the Presence of Latent Confounders with Skeleton Posterior*

Venue: ACM SIGKDD 2024

Authors: Pingchuan Ma et al.

Research organizations:

- HKUST
- Microsoft Research
- Shanghai Jiao Tong University

Problem

Scalable causal discovery with latent confounders.

State-of-the-Art Reference — SPOT (Continued)

Key innovation

Use a **skeleton posterior** rather than a single skeleton estimate to guide differentiable MAG optimization.

Reported findings

- Skeleton F1: **+8%**
- Arrowhead F1: **+13%**
- Tail F1: **+16%**
- SPOT outperformed its compared methods across most evaluation settings

Relationship to our project

SPOT is our selected recent SOTA algorithm and the primary method we will theoretically investigate, reproduce, and evaluate.

Initial KPIs

How Will We Measure Success?

Skeleton quality

- **F1 Score**
- Precision
- Recall / TPR
- False Discovery Rate

Causal structure quality

- Arrowhead F1
- Tail F1
- **Structural Hamming Distance (SHD)**

Posterior quality

- AUROC
- AUPRC
- KL divergence

Efficiency

- Runtime
- Scalability with number of variables

- SPOT itself evaluates skeleton, arrowhead and tail F1/TPR/FDR, and evaluates skeleton-posterior inference using AUROC, AUPRC and KL divergence.

Experimental Plan

Phase 1: Understanding & Reproduction

- Study SPOT implementation and mathematical formulation
- Reproduce the Sachs experiment
- Verify preprocessing and evaluation protocol

Phase 2: Baseline Comparison

- Run/evaluate ABIC
- Compare structural recovery against SPOT
- Include NOTEARS where technically meaningful as a DAG reference

Phase 3: Skeleton Posterior Analysis

- Evaluate $\mathbb{P}(S \mid D)$
- Compare posterior quality with alternative approaches
- Analyze uncertainty in edge predictions

Phase 4: Ablation / Robustness

- Posterior guidance vs no posterior guidance
- Vary sample size
- Investigate sensitivity to noise / confounding
- Analyze computational cost

Expected Contribution

What Do We Aim to Establish?

- Understand the role of **skeleton uncertainty** in causal discovery
- Reproduce the SPOT framework on a real benchmark
- Quantify the effect of posterior-guided optimization
- Compare SPOT with appropriate causal-discovery baselines
- Study robustness and scalability
- Identify limitations and possible improvements

Central hypothesis

Representing skeleton uncertainty probabilistically can provide more reliable guidance than committing to a single estimated skeleton.

Team Responsibilities

Member	Primary Responsibility
Anant Narayan	Literature Review + Data Preparation & Experiment Control
Ankit Kumar	ABIC Baseline Implementation
Milan Shah	SPOT Implementation
All Members	Integration, Testing, Evaluation & Analysis

Timeline

M1 — Scoping

→ Problem definition, literature, assumptions, KPIs

M2 — Implementation

→ Data pipeline + ABIC + SPOT

M3 — Integration

→ Complete Sachs benchmark + variable-removal experiments

Final

→ Evaluation + analysis + report + presentation

Final Takeaway

Final Takeaway

Problem

Latent confounders make causal discovery ambiguous.



Challenge

Differentiable optimization makes learning scalable, but does not resolve skeleton uncertainty.



SPOT

Estimates a **posterior over possible skeletons** $q\phi(S|X)$



Posterior-Guided Optimization

Uses this uncertainty to guide **MAG structure learning**



Evaluation

SPOT vs ABIC

under complete and incomplete observation

Bottom line: Can skeleton-posterior guidance improve causal discovery under latent confounding?

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Thank You!

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